

BASITH SALMAN

(917) 656-4593 | basithsalman@mindspring.com | Irving, TX | US Citizen (DoD NACI Cleared)

PROFESSIONAL SUMMARY

Principal Systems Architect & Lead Software Engineer with an M.S. in Computer Science and over two decades of enterprise engineering experience. Recognized expert across ultra-low-latency market data/trading systems, mission-critical distributed backend architectures, and high-volume retail/defense POS platforms. Proven track record leading multi-million-dollar modernization initiatives, optimizing sub-millisecond execution engines, and engineering full-stack web/API ecosystems across financial institutions, defense enterprises (DoD), and telecommunications providers.

CORE COMPETENCIES & TECHNICAL SKILLS

Architecture & Systems:	Ultra-Low Latency, High-Throughput Engines, Linux Kernel/OS Tuning, Microkernel Concepts, Distributed Systems, Point of Sale (POS) Architecture, IPC, Shared Memory, TCP/IP, SOCKS5.
Programming Languages:	Modern C++ (11/14/17), C, Python, PHP, Java, C#, Go, JavaScript, Bash/Ksh, SQL, Awk/Sed.
Web & Frameworks:	Modern PHP (Laravel, Symfony, Zend), Python (Django, Flask), React, Angular, jQuery, Node.js, REST APIs, Microservices, MVC Architecture, Bootstrap.
Data, Messaging & Ops:	MariaDB/MySQL, PostgreSQL, Oracle, Sybase, Redis, Kafka, MSMQ, RabbitMQ, Docker, AWS/EC2, Red Hat Enterprise Linux / Real-Time, Ubuntu, Windows Server.
Security & Compliance:	PKI, OpenSSL/TLS, HSM Integration, EMV Payment Processing, Root CA Security, DoD NACI Cleared.

PROFESSIONAL EXPERIENCE

Principal Software Consultant / Independent Engineer — Self-Employed 11/2023 – Present | Remote

- Architect and deliver custom financial stock screeners, low-latency market analysis tools, and scalable e-commerce platforms using Modern C++, Python, PHP (Laravel), React, and RESTful APIs.
- Design robust backend automations, high-throughput job queue systems, and secure transaction workflows for commercial clients.

Application Developer III (Senior Infrastructure Lead) — Ziply Fiber 03/2020 – 09/2023 | Remote

- Architected and maintained core backend infrastructure across 4 Northwestern states, managing critical outage tracking, incident workflows, and subscriber lifecycle management.
- Designed and implemented high-throughput REST APIs and backend services for Matrix 4/5 portals utilizing Modern C++, Python, MariaDB, and Laravel.
- Owned the mission-critical Serviceman authentication and subscription subsystem, managing Walled Garden flows, LDAP, and RADIUS integration.
- Maintained real-time network visualization mapping components interfacing low-level C++ services with dynamic web frontends based on CLLI network topology.

Lead POS & Full Stack Engineer — Army & Air Force Exchange Service (AAFES) 03/2019 – 03/2020 | Dallas, TX

- Engineered backend services and responsive modules for an \$8B DoD e-commerce portal (aafes.com) serving global military personnel using C++, PHP, and JavaScript.
- Built automated hardware integrations for point-of-origin weighing and real-time shipping logistics calculation.

Senior Infrastructure & Security Developer — JPMorgan Chase 11/2018 – 02/2019 | Lewisville, TX

- Engineered real-time synthetic monitoring and grey-failure detection engines in Modern C++ and .NET Core to protect enterprise Hardware Security Modules (HSMs) housing root CA keys.

Senior Software / POS Architect (Consultant) — Pizza Hut 03/2018 – 08/2018 | Plano, TX

- Designed core modules for restaurant POS systems utilizing C++/Qt, PHP, and Google mapping APIs to support beer delivery compliance, GPS logging, and dispatch automation.

Web Developer Consultant (Freelance Part-Time) — FMTC 03/2018 – 12/2018 | Austin, TX

- Engineered merchant API parsers and high-throughput link resolvers using Modern C++, Python, Laravel, Angular, and jQuery.

Senior Systems & Security Architect (Consultant) — Wells Fargo 03/2017 – 12/2017 | Charlotte, NC

- Architected key components of the enterprise Secure Software Portal (SSP) to orchestrate server analytics, system compliance, and automated backups across thousands of bank servers using C++, PHP (Zend), and C#.

Senior Systems Architect (POS Modernization) — NEC / 7-Eleven Inc.

05/2016 – 03/2017 | Irving, TX

- Architected the convergence of fuel and retail POS terminals across 800+ stores in North America using C/C++ and .NET, eliminating dual-system hardware footprints.

Senior Systems Architect (Enterprise Defense POS) — AAFES

08/2014 – 03/2016 | Dallas, TX

- Architected global EMV (Chip & PIN) payment processing infrastructure across worldwide military installations for a \$15B DoD enterprise.
- Formulated the architectural blueprint that catalyzed an \$80M modernization of military retail POS infrastructure.

Ultra-Low-Latency Trading Infrastructure Engineer — Morgan Stanley

03/2013 – 11/2013 | New York, NY

- Optimized institutional securities execution platform in Modern C++/C; achieved significant latency reductions via application profile refactoring and custom Linux kernel tuning.
- Executed the critical codebase consolidation reconciling Morgan Stanley and Smith Barney trading engines.

Lead Systems Architect / Trading Infrastructure Lead — NYSE Euronext

12/2011 – 03/2013 | New York, NY

- Directed the software infrastructure migration of the NYSE Universal Trading Platform (UTP) across hundreds of servers from NYC to Chicago data centers.
- Reduced end-to-end transaction latency by architecting a transition to Red Hat Enterprise Real-Time kernels, leading multi-million-dollar platform enhancements.

Assistant Vice President (AVP), Software Development — Tradeweb

12/2010 – 11/2011 | Jersey City, NJ

- Engineered multi-asset fixed-income trading software using high-performance C++/STL, supporting daily institutional trading volumes exceeding billions of dollars.

Senior Systems Engineer (Market Data Infrastructure) — Interactive Data Corp (IDC)

03/2007 – 10/2010 | White Plains, NY

- Developed sub-millisecond market data feed engines delivering real-time quotes to 80 countries worldwide using C/C++ and PHP.
- Diagnosed and resolved a multi-year Trans-Atlantic packet delivery latency bottleneck between NYSE and London market data endpoints.
- Built high-precision bandwidth burst recording systems underpinning \$90M in annual customer billing infrastructure.

EARLIER CAREER HISTORY (SELECTED TECHNICAL ENGAGEMENTS)

Senior Systems Software Engineer — Philips Medical Systems	2006 – 2007
Senior Security Software Architect — Permeo Technologies / NEC (Spun-off SOCKS5 Shim Layer Firewall)	Earlier Engagement
Operating Systems Consultant — IBM (Mach Microkernel Research & Infrastructure)	Earlier Engagement

EDUCATION & CREDENTIALS

Master of Science (M.S.) in Computer Science	University of Massachusetts Lowell, Lowell, MA
Bachelor of Science (B.S.) in Electrical & Electronics Engineering	National Institute of Technology (NIT), Surathkal, India
Security Clearance & Status: US Citizen DoD NACI Clearance Verified	